Integration of visual attributes in spatial segmentation based on temporal asynchrony revealed by a noise masking paradigm

Yen-Ju Chen[1, 2]
Zitang Sun[2]
Shin'ya Nishida[2]

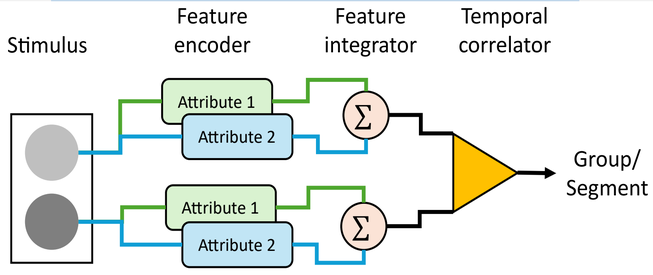
[1]Department of Psychology, Soochow University, Taipei, Taiwan
[2]Cognitive Informatics Lab, Graduate School of Informatics, Kyoto University, Kyoto, Japan

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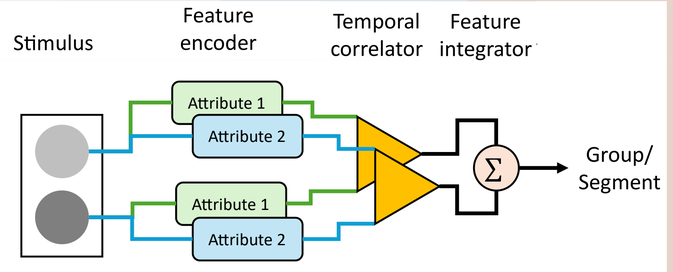
- Temporal correlation computation underlies asynchrony-based spatial segmentation (temporal segmentation) across different visual attributes (Chen et al., 2025).
- The visual system can simultaneously utilize multiple attributes to support temporal segmentation (Chen et al., 2024; Guttman et al., 2005; Blake & Lee, 2005).
- However, how these attributes are integrated remains unclear. We therefore propose two alternative computational architectures and test them using psychophysical experiments and computational modeling.

Hypothesis 1: Single channel for correlation computation



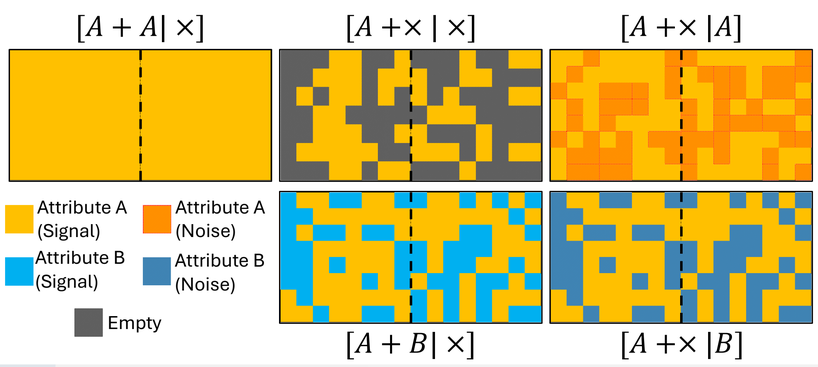
Cross-masking between attributes 1 & 2

Hypothesis 2: Multiple independent channels for correlation computation



No cross-masking

Spatial arrangement of different signal-noise combination



Notation:

[***| ***]

Segmentation signal

Temporally uncorrelated noise

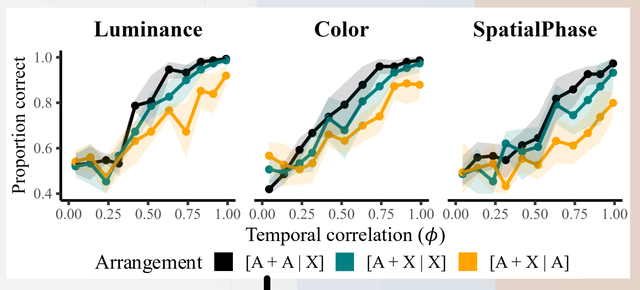
A, B = attribute type
X = absent

Condition:
[A+A|X] Full signal
[A+X|X] Half empty
[A+X|A] Within-attribute noise
[A+B|X] Cross-attribute signal
[A+X|B] Cross-attribute noise

Task:

- 2-IFC segmentation task: segmented vs. unsegmented lattice.
- Segmentation strength was determined by within-region temporal correlation (φ; 0–1, step = 0.1).
- Cross-region temporal correlation was fixed at 0.

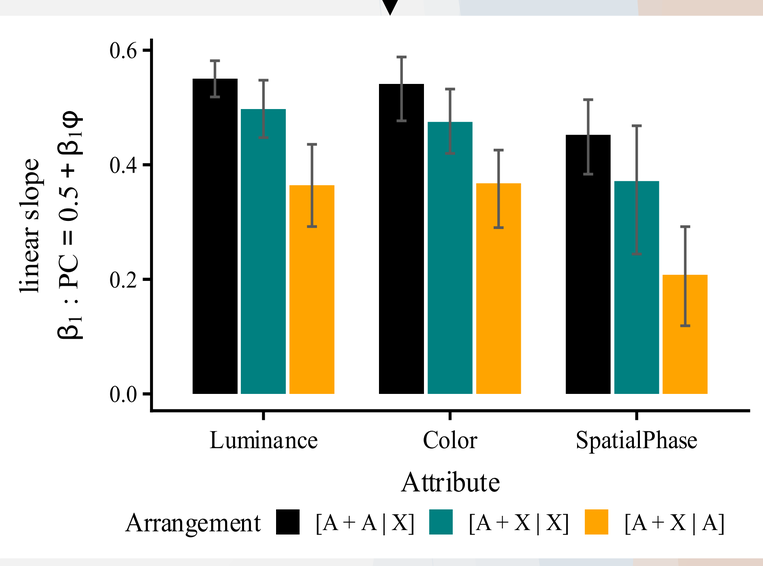
Exp 1. Single Attribute Experiment



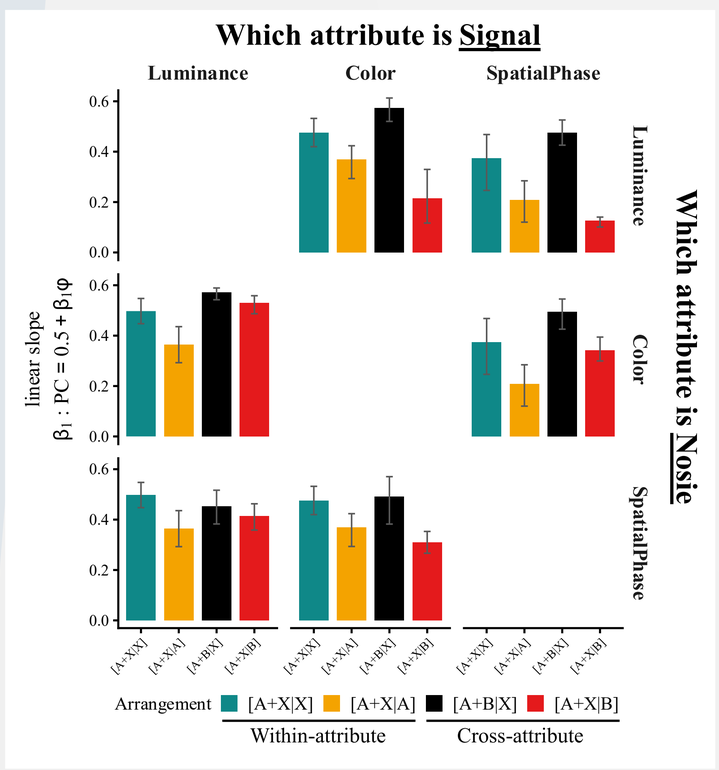
- Across all attributes, segmentation performance followed the pattern **[A + A |X]** > **[A + X |X]** > **[A + X | A]**
- The reduction from **[A + A | X]** to **[A + X | X]** indicates that temporal segmentation relies on spatial integration of local temporal correlation
- The additional reduction from **[A + X | X]** to **[A + X | A]** shows that temporally uncorrelated noise is actively incorporated into the correlation computation rather than being ignored
- Therefore, noise carried by the same attribute can disrupt the estimation of temporal correlation and impair segmentation performance

Linear regression with the function:

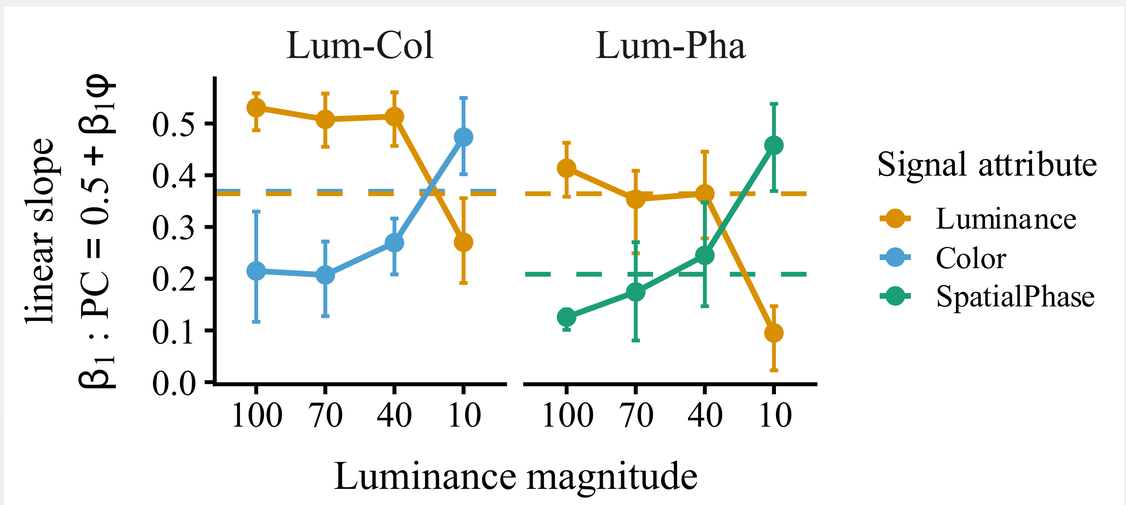
$$\boldsymbol{Proportion\ correct = 0.5 + \beta_1 \phi}$$



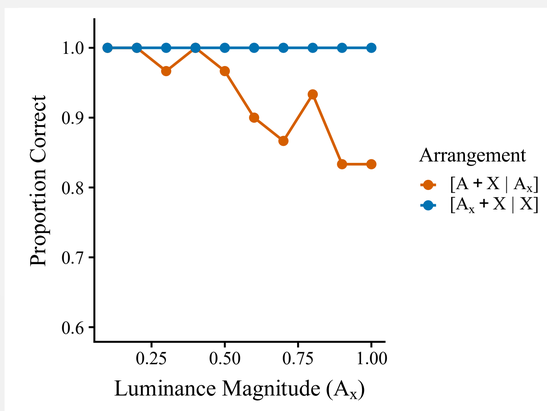
Exp2. Cross Attribute Experiment



- **[A + X | B]** ≤ [A + B | X] Cross-attribute noise reliably impaired temporal segmentation across attribute combinations
- In several conditions, cross-attribute interference approached within-attribute noise levels, suggesting interactions between attributes during temporal correlation computation.
- The interference was asymmetric: luminance >> color, spatial phase



- Cross-attribute interference systematically changed with luminance magnitude.
- Lowering luminance magnitude reduced its interference with color and spatial-phase segmentation, while increasing the ability of color and spatial-phase noise to disrupt luminance-based segmentation.
- These results suggest that interference is governed by relative signal strength across the attributes, supporting a gain-weighted integration mechanism.



Supplementary Experiment

- Temporal segmentation remained near ceiling even at 10% luminance magnitude.
- Thus, low stimulus energy was sufficient to drive temporal correlation computation.
- Performance dropped only when weak luminance signals were integrated with stronger luminance noise, suggesting a role of relative signal strength.

Model

Gain-weighted single-channel model

$$S, N = \begin{bmatrix} 1 \\ \boldsymbol{G_{col}} \\ \boldsymbol{G_{Pha}} \end{bmatrix}^T \begin{bmatrix} I_{Lum} \\ I_{Col} \\ I_{Pha} \end{bmatrix}$$

$$SNR = \frac{S}{\boldsymbol{\sigma_{inner}} + N}$$

$$\beta = A(1 - e^{-SNR})$$

$$P(\phi) = 1 - 0.5e^{-\beta\phi^2}$$

- Attributes are integrated before temporal correlation computation.
- Interference is determined by relative gain rather than independent channels.
- The model (G_{Col} = 0.33, G_{Pha} = 0.25, σ_{inner} = 0.15) explained 88% of the behavioral variance

Discussion

- Among luminance, color, and phase shift, spatial segmentation by temporal asynchrony in one attribute can be masked by temporal noise in another attribute.
- Luminance noise masks color or phase shift signal more effectively than vice versa under the full-strength (100% contrast) conditions, but the three attributes can be equally effective as a noise masker in cross-attribute masking when stimulus strengths are properly adjusted
- The results support Hypothesis 1

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